

Turn shared HVAC from a hidden trade-off into a visible, governable business decision.

EcoHVAC Guardian helps decision-makers test how intelligent coordination could improve occupant comfort, reduce disruption and create measurable value—before increasing operational authority.

THE OPPORTUNITY

- | | | |
|----|--------|--|
| 01 | VALUE | Focus investment on measurable energy and reliability outcomes |
| 02 | PEOPLE | Make comfort and service fairness visible |
| 03 | TRUST | Expand capability only as evidence and safeguards mature |

Recommendation Use a governed evidence pilot to validate value, protect people and earn the next stage.

A shared AHU turns separate room demands into one comfort, reliability and cost decision.

When capacity is constrained, local room requests create cross-zone consequences that conventional dashboards can leave hidden.



- COMFORT**
Which occupied rooms receive scarce airflow?
- RELIABILITY**
Can degradation and future risk be surfaced early?
- COST**
Can avoidable energy and disruption be reduced?
- BUSINESS NEED**
One decision view across four room twins and one 0.24 m³/s AHU

The modeled value case lives or dies on avoided incidents—not energy savings alone.

The commercial opportunity is visible, but the range from no payback to 8.1 months makes incident evidence the pilot’s first management question.

ILLUSTRATIVE BASE CASE

\$9,080/year

Annual net value · 33.04-month payback

Energy value	+ \$1,080
Avoided incidents	+ \$12,000
Annual support	– \$4,000

SENSITIVITY

LOW	–\$1,600	No payback
BASE	\$9,080	33.0 months
HIGH	\$29,625	8.1 months

≈92% of gross modeled benefit comes from avoided incidents.

BUSINESS EVIDENCE GATE

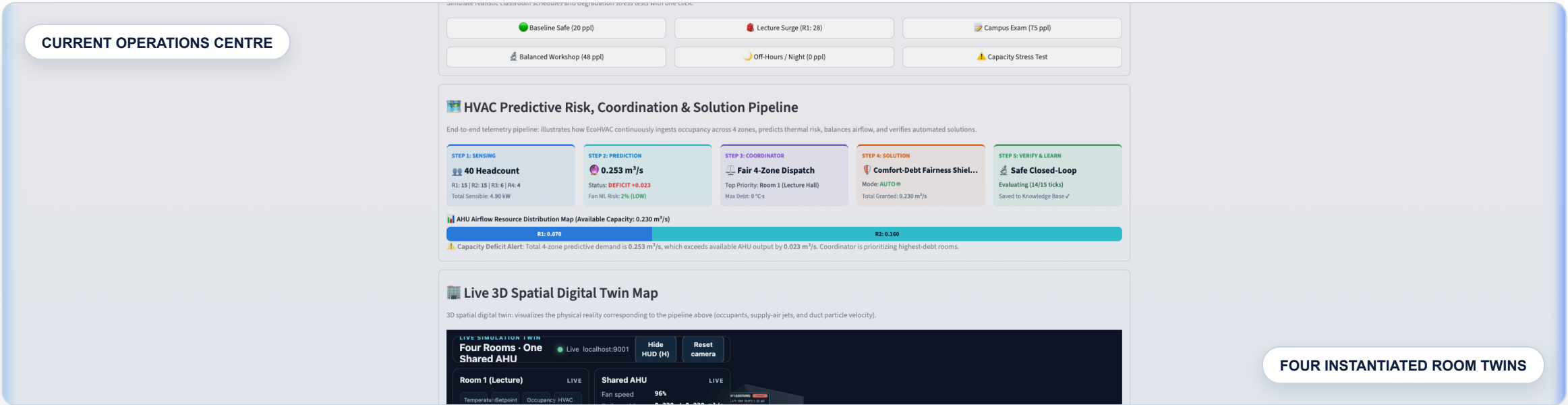
Replace the key assumptions with measured evidence.

- Normalized HVAC energy baseline
- Incident frequency, severity and avoidability
- Applicable tariff and supplier costs
- Comfort and operator-workload outcomes

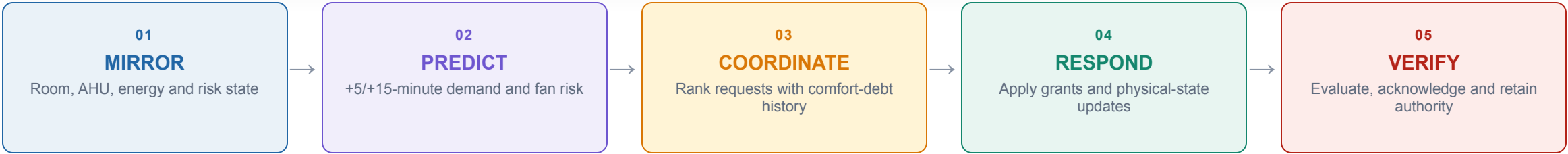
Editable illustrative scenarios—not measured savings, forecasts, quotations or an approved investment.

The Digital Twin converts a shared constraint into a closed operating loop.

The interface is the window; the differentiator is persistent state that connects prediction, coordination, simulated response and verification.



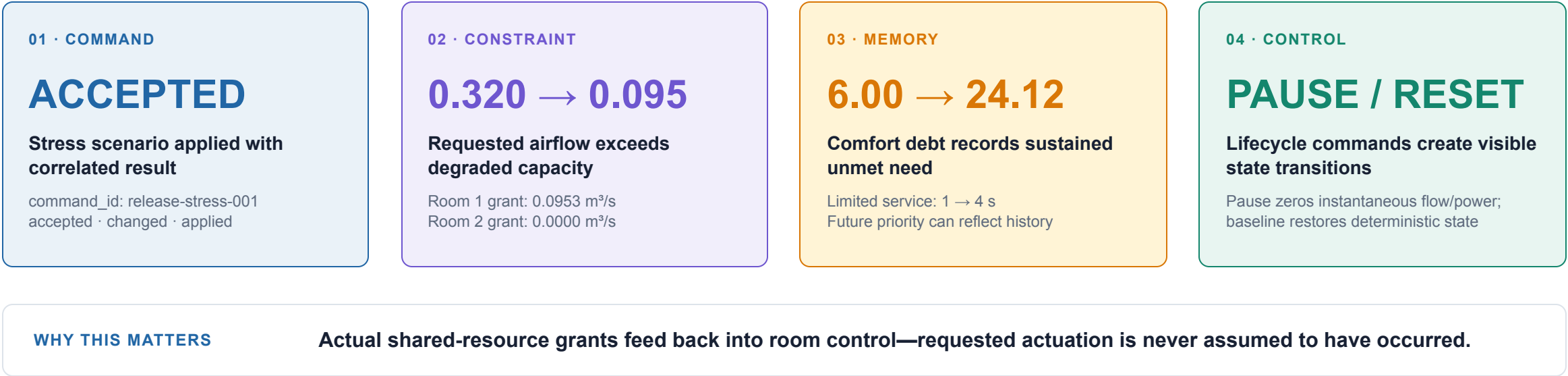
FOUR INSTANTIATED ROOM TWINS



Current UI/state evidence. Room 3/4 MQTT command/status plumbing is not yet symmetric with Rooms 1/2.

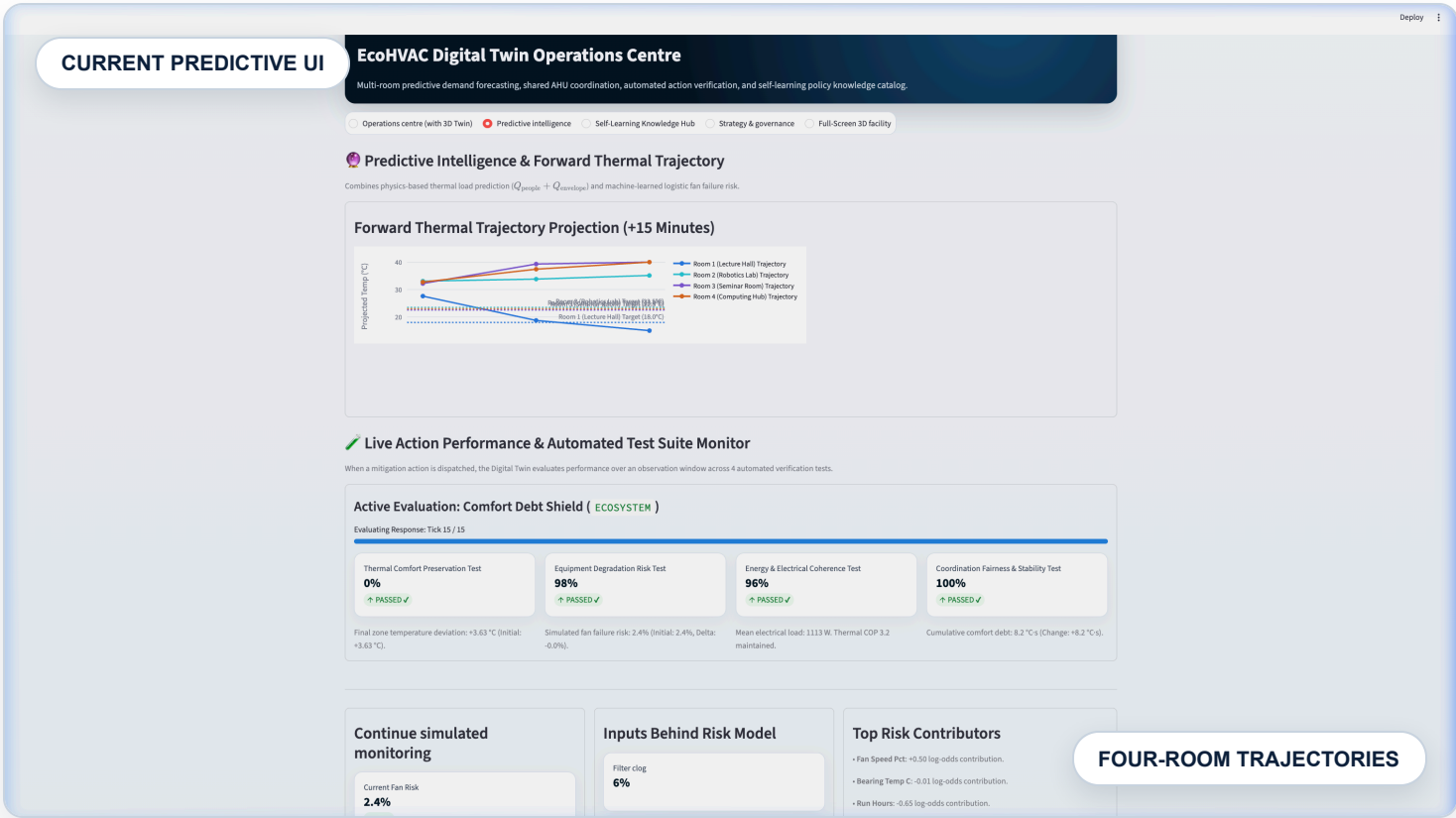
The twin records what airflow was actually granted—not merely what each room requested.

An archived two-room integration trace shows causal allocation, actuator feedback and bounded memory of unmet service.



The twin projects what may happen next—and abstains outside supported inputs.

Thermal trajectories, equipment-risk evidence and post-action evaluation are connected to the same operating state.



+5 / +15 min

Interpretable continuation projections by room

OOD → ABSTAIN

Missing, invalid or unsupported inputs do not become “low risk”

15 ticks

Post-action checks for comfort, risk, energy and stability

161 tests

Software evidence across control, fairness, energy, model and messaging

Fan-risk evidence uses synthetic training/holdout data; trajectories are deterministic teaching-model projections, not field forecasts.

Trust is an operating function: expose who waits, preserve human authority and secure every step.

The twin’s value depends on making burdens, decision rights and connection risks visible—not treating them as deployment footnotes.

PEOPLE

Benefits and burdens share one ledger.

- Comfort deviation and denied-flow duration
- Bounded comfort debt records unmet service history
- Aggregate occupancy only—never identity or profiling
- Missed-risk and operator-workload review

AUTHORITY

People retain high-impact decisions.

- Correlated application acknowledgement
- Pause, resume and simulation emergency stop
- Human approval or rejection of candidate policy
- Named authority, stop rule and rollback before expansion

SECURITY

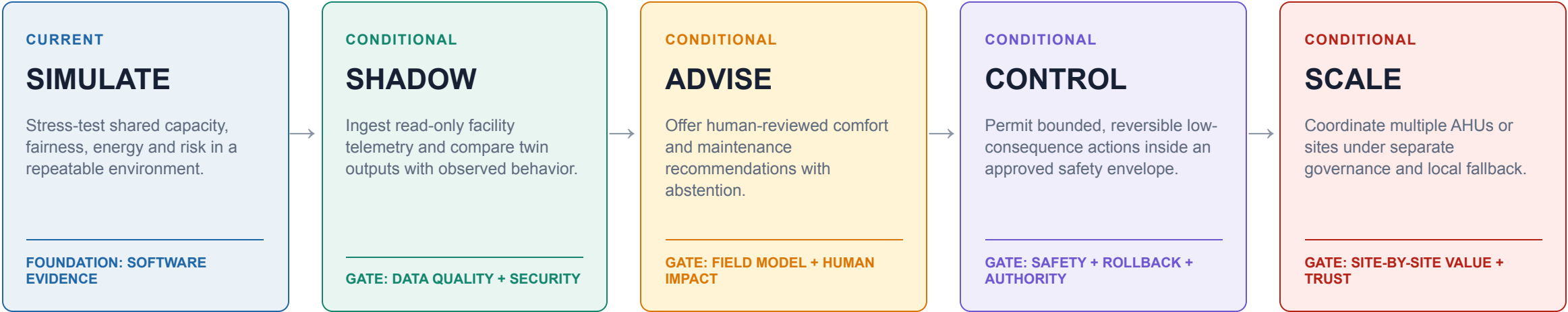
Connectivity must be earned.

- Current: strict validation, replay discipline, local audit option
- Current broker: anonymous MQTT/WebSockets in plaintext
- Before connection: identity, TLS/WSS and least privilege
- Protected evidence, fallback and rehearsed recovery

Synthetic risk evidence is not a field failure rate; hardened broker controls are deployment targets, not current operating evidence.

Operational authority can expand only as evidence, safeguards and accountability mature.

The Digital Twin creates a staged path from classroom simulation to separately governed site scale—without jumping directly to autonomy.

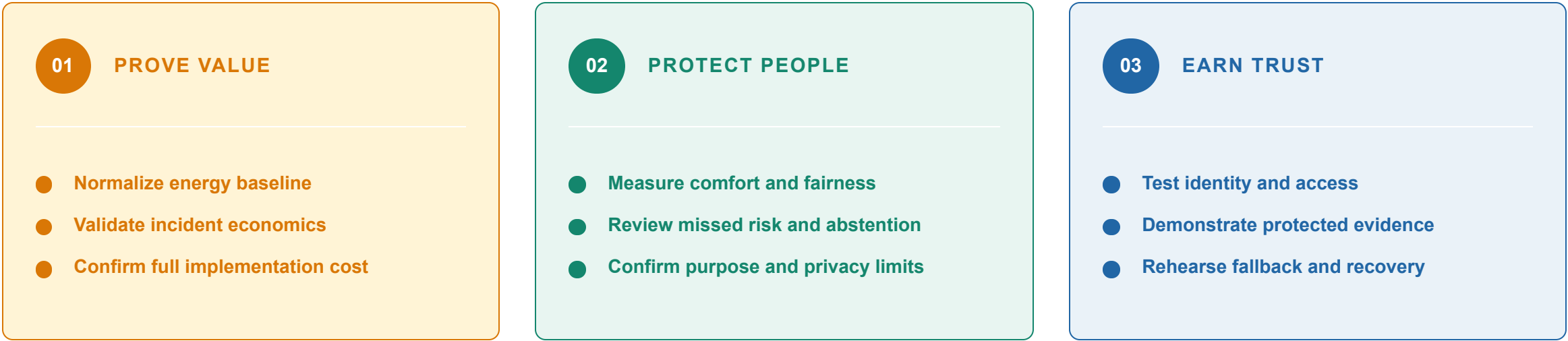


POTENTIAL VALUE Move from understanding one shared asset to learning where predictive coordination creates repeatable value—while preserving local human control.

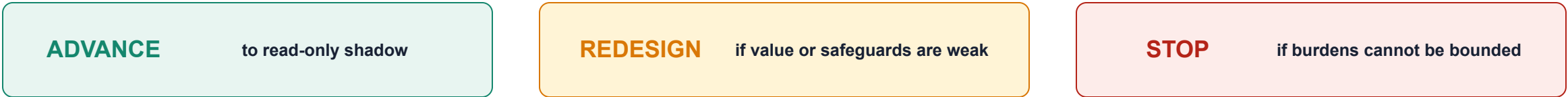
Future stages are proposed and evidence-gated; no schedule, budget, facility approval, physical autonomy or site-scale benefit is asserted.

Approve an evidence-building pilot—not autonomous deployment.

Use the Digital Twin to prove value, protect people and earn the evidence required for read-only shadow operation.



EVIDENCE-BASED EXIT



Current foundation: four-room simulation, one shared AHU, predictive intelligence, governed command lifecycle and 161 passing software tests.

Decision scope: classroom simulation pilot; no facility authorization, physical autonomy, budget, schedule or signatory is asserted.